

Offline rollout materialization

Factual state and finite table

$$s_t^{\text{S0-pose}}, b_t, Q_t$$
$$m_{t,i}^{\text{act}}, \rho_{t,i}$$

Target RRI oracle

r_t^e for finite valid rows
label and support masks

One factual selected action

$$a_t \in A_t$$

successor: pose, history, budget, lineage

Replay store

q_h/: *state, masks, labels*,
selected transition, provenance

persisted q_h/ replay

Offline fitted-Q learning

Causal batch

$$(s_t, Q_t) \text{ and } (s_{t+1}, Q_{t+1})$$

only factual successor support

Fitted-Q objective

$h = 1$: all finite hard-valid labels
 $h > 1$: $Q_h \rightarrow Q_{h-1}$ on selected support

Optional Double-Q ablation

online scorer selects; delayed scorer evaluates
not the definition of Q_H

Held-out policy evaluation

π_Q selects only hard-valid rows
oracle rescoreing measures endpoint gain